



SALAS O'BRIEN
3200 WINDY HILL ROAD SE
SUITE 200E
ATLANTA, GEORGIA 30339
PHONE 404-881-5300

Project: Wehunt Preserve; Davidson Homes
Location: Braselton, GA
Description: Ash - B, Slab, CBonus, ACH50

Date: August 8, 2024
By: JTH / TJH
House Faces: Worst NorthEast

These Loads are dependent on the information/assumptions listed below. **Any deviation from these assumptions during construction could have a significant impact on the cooling and heating loads for this house and render the recommended equipment sizing invalid.** If any of these assumptions are incorrect, please let us know immediately so we can re-calculate them. This information will be used in a residential heating and cooling load program based on calculations recommended by ASHRAE, Handbook of Fundamentals. Equipment sizing indicated in this report is valid only upon verification that actual construction is in full compliance with the descriptions below and independent testing of HVAC systems has been performed; such verification and testing shall be performed by Vallus.

BUILDING DATA	
Typical Ceiling Height	9.0 ft
Floor Area	2,459 ft ²
Volume	21,832 ft ³
Percentage Glass	11 %
Number of Bedrooms	3

DESIGN CONDITIONS		
	Summer	Winter
Outdoor	95 °Fdb	18 °Fdb
Indoor	75 °Fdb	72 °Fdb
ΔT	20 °Fdb	54 °Fdb
Range	17.3 °Fdb	

ENERGY CALCULATIONS	
Right Choice HVAC kWh	8891
SEER	14
ACH50	3.0
Mechanical Ventilation	YES
House Outside Air CFM	60

CONSTRUCTION DESCRIPTIONS						
Walls / Partitions	#	U-Value	Description/Notes			
	W1	0.077	Above Grade	2x4	R-13 batt	
	W2	0.067	Basement	Concrete + 2x4	R-13 batt	
	W3	0.053	Attic	2x6	R-19 batt	
Doors	#	U-Value	Description/Notes			
	D1	0.13	Exterior Door		R-7.7	
	D2	0.083	Garage Door		R-12	
Roofs	#	U-Value	Description/Notes			
	R1	0.026	Flat Ceiling		R-38 blown	
	R2	0.026	Vaulted		R-38 batt	
Floors	#	U-Value	Description/Notes			
	F1	0.026	Framed		R-38 batt	
	F2	0.1	Slab		None	
Glass	#	Glass Type	Windows	U-Value	Internal Shading	Notes
	G1	Double Insulated	All types	0.35	Blinds/Draperies	SHGC 0.22
	G2					
	G3					

LOAD SUMMARY					
		Unit 1 - Whole House			
		Cooling		Heating	
Sensible Load		25866 Btu/hr		29238 Btu/hr	
Minimum Nominal Capacity		2.87 Tons		8.57 kW	
Load Density		856 SF/Ton		3.5 W/SF	
Floor Area Served		2459.0 SF		2459.0 SF	
Mechanical Ventilation		60 CFM		60 CFM	
System Size		3.0 Tons		10.0 kW	
Leaving Air Temperature		60.1 deg F		92.6 deg F	
Installed Capacity		820 SF/Ton		4.1 W/SF	

I understand the information provided herein, and warrant, to the best of my ability and knowledge, and to the full extent of my control as the Contractor, that this house will be built in accordance with the information given above.

Signed _____ Title _____ Date _____

Project:	Wehunt Preserve; Davidson Homes
Location:	Braselton, GA
Description:	Ash - B, Slab, CBonus, ACH50

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[illegible]

Unit 1

Size

3.00

Tons

Airflow

1200

CFM

Heat kW

10

kW

First Floor Airflows

Room	CFM Cool	CFM Heat	CFM Avg	Duct Size
Foyer	24	41	32	
Dining	69	132	100	
Kitchen	166	92	129	
Breakfast	72	103	88	
Family	180	138	159	
Owner's Suite	116	136	126	
Owner's Bath	28	52	40	
Owner's WIC	24	39	32	
Laundry	48	16	32	
Mud Rm	12	16	14	
Powder	4	7	6	
-	-	-	-	
-	-	-	-	
-	-	-	-	

[illegible]

Cooling Size	2	DX Cooling kWh	2139
Heating Size	2	RC Heating kWh	3499
Best Case	2	Total kWh	5638

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DX Cooling kWh	1310
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By: JTH / TJH

RC Heating kWh	1943
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Level(s): Second Floor

Total kWh	3253
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SEER 14

COOLING LOAD CALCULATION						Ceiling Height		18		8		8		8		8		8		8		8		8		8		8									
Components						CLTD (°F)		U-Value (Btu/hr-sq-ft-F)		Cooling Load Factor		Entire House		Stairs		Hallway & Alcove		Bed 2		Bath		Bed 3		Bonus													
	Type	Description				Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr	Qty	Btu/hr		
Glass	G1	NorthWest	Right	(E)	18	Btu/hr-sf	30	552		-		-		-		-	30	552		-		-		-		-		-		-		-		-		-	
	G1	North		(SE)	6	Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-	
	G1	NorthEast	Front	(S)	8	Btu/hr-sf	42	329		-	12	94	30	235		-		-		-		-		-		-		-		-		-		-		-	
	G1	East		(SW)	13	Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-	
	G1	SouthEast	Left	(W)	14	Btu/hr-sf	15	214		-		-		-	15	214		-		-		-		-		-		-		-		-		-		-	
	G1	South		(NW)	16	Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-	
	G1	SouthWest	Back	(N)	26	Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-	
	G1	West		(NE)	29	Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-	
	G1	Sky Light			46	Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-	
G1	Shaded			6	Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		-		
Door	D1	Door		21	0.13	2.73	Btu/hr-sf	0	0		-		-		-		-		-		-		-		-		-		-		-		-		-		
	D2	Door		55	0.08	4.57	Btu/hr-sf	18	81		-	17.8	81		-		-		-		-		-		-		-		-		-		-		-		
Above Grade Walls	W1	NorthWest		19	0.08	1.46	Btu/hr-sf	205	300	-	-	34	50	18	26	-	-	-	-	152	223	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	W1	North		13	0.08	1.00	Btu/hr-sf	0	0	-	-		-		-	-	-		-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W1	NorthEast		19	0.08	1.46	Btu/hr-sf	218	319	-	-	42	61	84	123	-	-	-	-	92	134	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W1	East		23	0.08	1.77	Btu/hr-sf	0	0	-	-		-		-	-	-		-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W1	SouthEast		21	0.08	1.62	Btu/hr-sf	287	464	-	-	-	-	124	200	52	84	112	181	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W1	South		16	0.08	1.23	Btu/hr-sf	0	0	-	-		-		-	-	-		-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-		
	W1	SouthWest		21	0.08	1.62	Btu/hr-sf	0	0	-	-		-	</																							